

D502

Data Analytics Capstone

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A regional credit union with 38 branches reports that member attrition rose from 6% to 9% last year. Its branches were remodeled in 2023. Which capstone research question best serves the need?

- A. Did the 2023 remodel cause members to leave?
- B. Which member behaviors and account features predict attrition within 12 months?
- C. How has the credit union industry evolved since 1950?
- D. How can the credit union eliminate all attrition?

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A capstone student at a hospital wants to test whether average ER wait time differs between its two ER sites. Waits are recorded in minutes for 3,000 visits. The hospital's CFO sponsors the project. Which test fits?

- A. A one-sample t-test against a 30-minute goal.
- B. A two-sample (independent) t-test.
- C. A chi-square test of independence.
- D. A paired t-test.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A capstone asks whether customer region (North, South, East, West) is associated with plan type (basic or premium). Which test fits?

- A. No test, since associations between categories cannot be tested.
- B. A one-way ANOVA.
- C. A chi-square test of independence.
- D. A two-sample t-test.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A capstone proposal for a staffing agency with 200 recruiters plans to predict placement success from candidates' social media posts, which the agency does not collect and has no consent to use. The agency's website was just redesigned. What is the main problem?

- A. The chosen model type is too simple.
- B. The website redesign will change the data.
- C. Nothing; any data posted online is always fair to use.
- D. The needed data is not available or permitted; rescope to data the agency holds and may use.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Which capstone proposal is most feasible?

- A. Predict turnover from data the student plans to collect over three years.
- B. Model all employee emails without an approval.
- C. Solve every HR problem with a single model.
- D. Predict 90-day employee turnover from the company's existing HR records, approved for this use.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A capstone data set from a clinic chain with 16 sites includes patient names, birth dates, and diagnosis codes. The student's laptop is encrypted. Before analysis, what should be done?

- A. Remove the diagnosis codes but keep the names.
- B. Remove or mask direct identifiers and get the required approval for the data use.
- C. Nothing more, since the laptop is encrypted.
- D. Publish the full data set in the appendix for transparency.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A capstone model forecasts weekly sales for a garden-supply store with 3 locations. On four test weeks the errors (actual minus forecast, in units) are +5, −3, +1, and −1. The owner started the business in 2010. What is the RMSE?

- A. 2.5
- B. 0.5
- C. 9
- D. 3

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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8. [TRANSFER — reversed stem / different variable]

A regression model's RMSE is 4. What is its MSE?

- A. 16
- B. 8
- C. 2
- D. 4

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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9.

A capstone finds that customers who use the mobile app spend 30% more. The data covers one year at a pet-supply retailer with 55 stores, and customers chose whether to install the app. The report was due in April. What limitation must the report state?

- A. Self-selection: app users may differ from others, so the result shows association, not that the app causes higher spending.
- B. The app causes 30% more spending.
- C. 55 stores is too few to report any finding.
- D. There are no limitations, because a full year of data is always enough.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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10.

A capstone for a gym chain with 12 clubs finds that members who visit fewer than 4 times in their first month are 3 times as likely to cancel by month six. The chain's CEO likes new equipment. Which recommendation follows from the finding?

- A. Buy new equipment for all clubs.
- B. Cancel any member who does not visit 4 times.
- C. Reach out to new members with fewer than 4 first-month visits (for example, coaching check-ins) and track cancellations.
- D. Report the finding with no recommendation.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A student plans a two-sample t-test on order values from two web stores, 40 orders each. The data is heavily skewed with extreme outliers. The stores sell outdoor gear. What should come first?

- A. Check the assumptions; with heavy skew and small samples, use a transformation or a nonparametric test such as Mann-Whitney U.
- B. Run the t-test anyway; it always works.
- C. Delete every order above the mean.
- D. Use a chi-square test instead.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Which assumption does linear regression make about its residuals?

- A. Residuals must all equal zero.
- B. Every predictor must be normally distributed.
- C. Roughly constant variance and independence, with no strong pattern against fitted values.
- D. Residuals should grow as fitted values grow.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A regional credit union ~~with 38 branches~~ reports that member attrition rose from 6% to 9% last year. ~~Its branches were remodeled in 2023.~~ Which capstone research question best serves the need?

Target word: best research question **Decides:** The need is rising attrition.

A	Camouflage	Chases an incidental detail with no evidence it matters.
B	Correct	Targets the problem and can be answered with member data.
C	Adjacent category	A history question, not the organization's need.
D	Absolute language	Not measurable or realistic.

2. Answer B Concept 2

A capstone student at a hospital wants to test whether average ER wait time differs between its two ER sites. Waits are recorded in minutes for 3,000 visits. ~~The hospital's CFO sponsors the project.~~ Which test fits?

Target word: statistical test **Decides:** Numeric outcome, two independent groups.

A	Dropped element	Drops the comparison between the two sites.
B	Correct	Compares mean wait time between two separate groups.
C	Confusion set	For two categorical variables, not a numeric mean.
D	Adjacent category	For two measures on the same subjects.

3. Answer C Concept 2 · transfer

A capstone asks whether customer region (North, South, East, West) is associated with plan type (basic or premium). Which test fits?

Target word: test for two categorical variables **Decides:** Both variables are categorical.

A	Absolute language	Chi-square does exactly this.
B	Adjacent category	Compares numeric means across groups.
C	Correct	Tests association between two categorical variables.
D	Confusion set	Needs a numeric outcome and two groups.

4. Answer D Concept 3

A capstone proposal for a staffing agency ~~with 200 recruiters~~ plans to predict placement success from candidates' social media posts, which the agency does not collect and has no consent to use. ~~The agency's website was just redesigned.~~ What is the main problem?

Target word: feasibility flaw **Decides:** The data is neither held nor permitted.

A	Adjacent category	The model does not matter without data.
B	Camouflage	Unrelated to the missing data.
C	Absolute language	Consent and terms still apply.
D	Correct	A plan without usable data is not feasible.

5. Answer D Concept 3 • transfer

Which capstone proposal is most feasible?

Target word: pick the feasible plan — reverse **Decides:** Existing, approved data.

A	Confusion set	The data will not exist within the project timeline.
B	Dropped element	Drops the permission requirement.
C	Absolute language	Unbounded scope.
D	Correct	The data exists and may be used.

6. Answer B Concept 4

A capstone data set from a clinic chain ~~with 16 sites~~ includes patient names, birth dates, and diagnosis codes. The student's laptop is encrypted. Before analysis, what should be done?

Target word: privacy step **Decides:** Direct identifiers are present.

A	Inverted operation	Removes the useful data and keeps the identifiers.
B	Correct	De-identification and approval come before analysis.
C	Camouflage	Encryption protects a lost laptop, not the analysis use.
D	Adjacent category	Exposes personal data.

7. Answer D Concept 5

A capstone model forecasts weekly sales for a garden-supply store ~~with 3 locations~~. On four test weeks the errors (actual minus forecast, in units) are +5, -3, +1, and -1. ~~The owner started the business in 2010.~~ What is the RMSE?

Target word: RMSE **Decides:** Squares: 25, 9, 1, 1; mean = 9.

A	Confusion set	That is the MAE: $(5 + 3 + 1 + 1) / 4$.
B	Adjacent category	The mean signed error; positives and negatives cancel.
C	Dropped element	The MSE, before the square root.
D	Correct	$\sqrt{9} = 3$.

8. Answer A Concept 5 • transfer

A regression model's RMSE is 4. What is its MSE?

Target word: solve for MSE **Decides:** $MSE = RMSE^2$.

A	Correct	$4^2 = 16$.
B	Confusion set	Doubles instead of squaring.
C	Inverted operation	Takes the square root instead of squaring.
D	Dropped element	Leaves out the squaring.

9. Answer A Concept 6

A capstone finds that customers who use the mobile app spend 30% more. The data covers one year at a pet-supply retailer ~~with 55 stores~~, and customers chose whether to install the app. ~~The report was due in April.~~ What limitation must the report state?

Target word: key limitation **Decides:** Customers chose to install the app.

A	Correct	Without random assignment, causation is not shown.
B	Confusion set	A causal claim, not a limitation.
C	Wrong stem number	Store count is not the main threat here.
D	Absolute language	Time span does not remove selection bias.

10. Answer C Concept 7

A capstone for a gym chain ~~with 12 clubs~~ finds that members who visit fewer than 4 times in their first month are 3 times as likely to cancel by month six. ~~The chain's CEO likes new equipment.~~ Which recommendation follows from the finding?

Target word: recommendation **Decides:** Low first-month visits predict cancellation.

A	Camouflage	Matches the CEO's preference, not the finding.
B	Absolute language	Turns a risk signal into a harsh rule.
C	Correct	Acts on the finding and measures the result.
D	Dropped element	Drops the action step.

11. Answer A Concept 8

A student plans a two-sample t-test on order values from two web stores, 40 orders each. The data is heavily skewed with extreme outliers. ~~The stores sell outdoor gear.~~ What should come first?

Target word: first step **Decides:** Small, skewed samples.

A	Correct	Assumption checks decide the right test.
B	Absolute language	Its validity depends on assumptions.
C	Dropped element	Removes real data.
D	Confusion set	Chi-square is for categorical data.

12. Answer C Concept 8 • transfer

Which assumption does linear regression make about its residuals?

Target word: assumption statement **Decides:** Constant variance and independence.

A	Absolute language	Some error is expected.
B	Confusion set	Normality concerns residuals, not predictors.
C	Correct	Homoscedastic, independent residuals are core assumptions.
D	Inverted operation	That describes a violation.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	3	
Absolute language	Always, never, guarantee — where judgment applies.	8	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Question tied to the organizational need	The research question must address the organization's stated problem and be answerable with its data.	Choosing an interesting but unrelated question, or an unmeasurable goal.	A business problem followed by candidate research questions.
2	Matching the test to the data	Two-sample t-test compares a numeric mean across two independent groups; chi-square tests association between two categorical variables; regression models a numeric outcome from predictors.	Picking a test by habit instead of by variable types.	The outcome type and the number of groups.
3	Feasibility and data access	A proposal is feasible only if the needed data exists, is accessible in time, and may be used for this purpose.	Planning around data the organization does not have or may not use.	Data that is not collected, not consented, or not yet available.
4	Privacy and de-identification	Remove or mask direct identifiers and get the required approval before analyzing personal data.	Believing device encryption alone makes identifiable data safe to analyze.	Names, birth dates, or record numbers in the data set.
5	Error metrics for numeric predictions	MAE = mean of absolute errors; MSE = mean of squared errors; RMSE = square root of MSE, in the original units and more sensitive to large errors.	Reporting MSE as RMSE, or averaging signed errors.	A list of errors and "What is the RMSE?"
6	Stating limitations	The report must name threats to validity, such as self-selection, which limit claims to association rather than cause.	Writing a causal conclusion from observational data.	Groups that chose their own treatment.
7	Actionable recommendations	Recommendations should follow directly from the findings and include a way to measure their effect.	Recommending what a stakeholder already wants, or stopping at the finding.	A clear finding and a list of possible actions.
8	Checking test assumptions	Check assumptions (independence, distribution shape, equal variance) before a parametric test; use transformations or nonparametric tests when they fail.	Running a t-test on small, heavily skewed samples without checking.	"Skewed," "outliers," or "small samples" next to a planned test.

Course scope (public description): The Data Analytics Undergraduate Capstone challenges students to demonstrate competencies supporting all BSDA program outcomes. Students will identify an organizational need, plan and develop a data analytics product to serve that need, and document the process in a project proposal and data project report.